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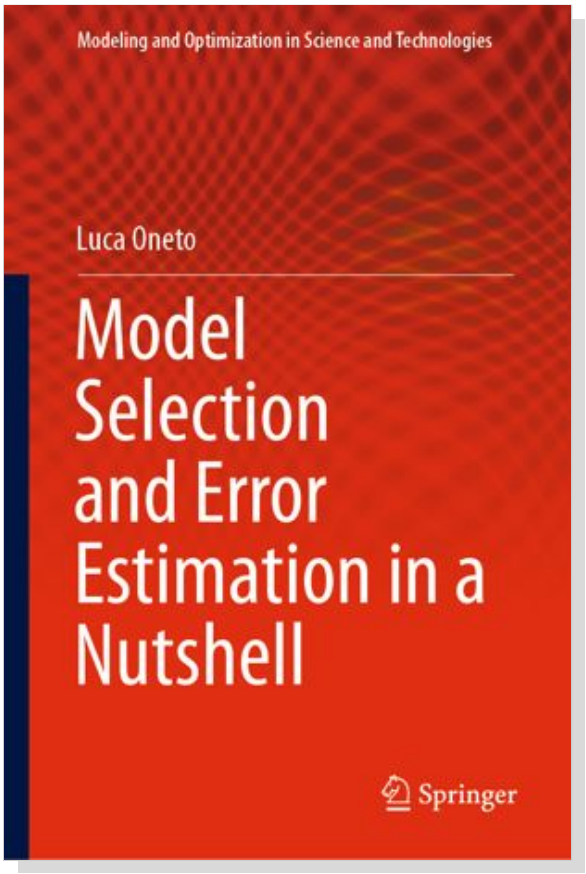
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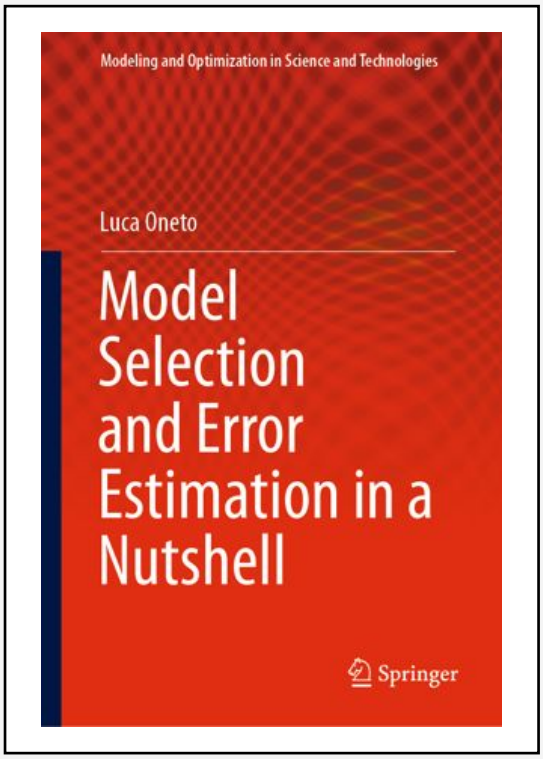
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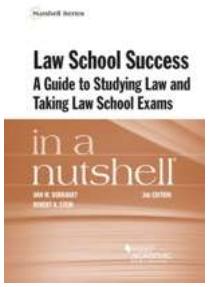
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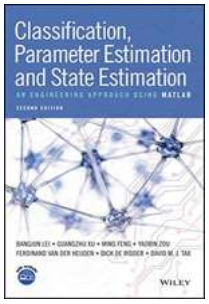


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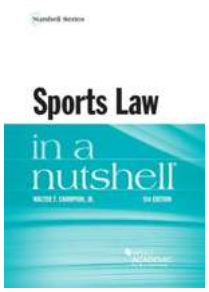
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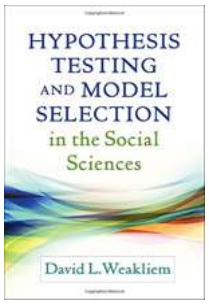
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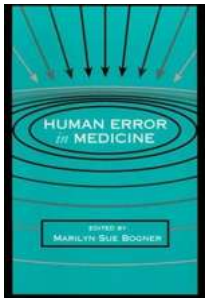
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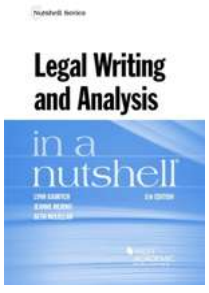
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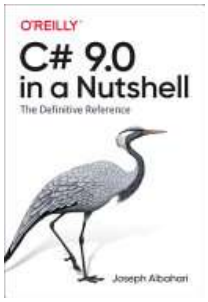
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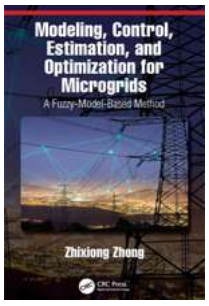
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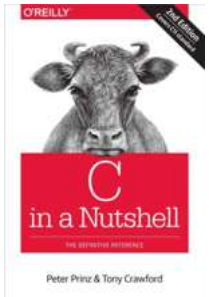
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Model Selection and Error Estimation in a Nutshell

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ISSN 2196-7326 ISSN 2196-7334 (electronic)
Modeling and Optimization in Science and Technologies
ISBN 978-3-030-24358-6 ISBN 978-3-030-24359-3 (eBook)
https://doi.org/10.1007/978-3-030-24359-3

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Do not worry Dad.
You know.
I understood you.
Even your mistakes.
And at the end, I am like you.
Luca Oneto

Foreword

The field of Machine Learning plays an increasingly important role in science and engineering. Over the past two decades, the availability of powerful computing resources has opened the door to synergistic interactions between empirical and theoretical studies of Machine Learning, showing the value of the "learning from example" paradigm in a wide variety of applications. Nevertheless, in many such practical scenarios, the performance of many algorithms often depends crucially on manually engineered features or hyperparameter settings, which often make the difference between bad and state-of-the-art performance. Tools from statistical learning theory allow to estimate the statistical performance of learning algorithm and provide a means to better understand the factors that influence algorithms behavior, ultimately suggesting ways to improve the algorithms or designing novel ones. This book reviews in an intelligible and synthetic way the problem of tuning and assessing the performance of an algorithm allowing both young researches and experienced data scientists to gain a broad overview of the key problems underlying model selection, state-of-the-art solutions, and key open questions.

London, UK
May 2019

Prof. Massimiliano Pontil

Massimiliano Pontil is a Senior Researcher at Istituto Italiano di Tecnologia, where he leads the CSML research group, and a part-time Professor of Computational Statistics and Machine Learning in the Department of Computer Science at University College London. He received a Ph.D. in Physics from the University of Genoa in 1999 (Advisor Prof. A. Verri). His research interests are in the areas of machine learning, with a focus on statistical learning theory, kernel methods, multitask and transfer learning, online learning, learning over graphs, and sparsity regularization. He also has some interests in approximation theory, numerical optimization, and statistical estimation, and he has pursued machine learning applications arising in computer vision, bioinformatics, and user modeling. Massimiliano has been on the program committee of the main machine learning conferences, including COLT, ICML, and NIPS, he is an Associate Editor of the *Machine Learning Journal*, an Action Editor for the *Journal of Machine Learning Research* and he is on the Scientific Advisory Board of the Max Planck Institute for Intelligent Systems Germany. Massimiliano received the Edoardo R. Caianiello award for the Best Italian Ph.D. Thesis on Connectionism in 2002, an EPSRC Advanced Research Fellowship in 2006–2011, and a Best Paper Runner Up from ICML in 2013.

Preface

How can we select the best performing data-driven model? How can we rigorously estimate its generalization error? Statistical Learning Theory (SLT) answers these questions by deriving non-asymptotic bounds on the generalization error of a model or, in other words, by upper bounding the true error of the learned model based just on quantities computed on the available data. However, for a long time, SLT has been considered only an abstract theoretical framework, useful for inspiring new learning approaches, but with limited applicability to practical problems. The purpose of this book is to give an intelligible overview of the problems of Model Selection (MS) and Error Estimation (EE), by focusing on the ideas behind the different SLT-based approaches and simplifying most of the technical aspects with the purpose of making them more accessible and usable in practice. We will start by presenting the seminal works of the 80s until the most recent results, then discuss open problems and finally outline future directions of this field of research.

Genoa, Italy
May 2019

Prof. Luca Oneto

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About the Author

Luca Oneto was born in Rapallo, Italy in 1986. He received his B.Sc. and M.Sc. in Electronic Engineering at the University of Genoa, Italy, respectively, in 2008 and 2010. In 2014, he received his Ph.D. from the same university in the School of Sciences and Technologies for Knowledge and Information Retrieval with the thesis "Learning Based On Empirical Data". In 2017, he obtained the Italian National Scientific Qualification for the role of Associate Professor in Computer Engineering and in 2018, he obtained the one in Computer Science. He worked as an Assistant Professor in Computer Engineering at University of Genoa from 2016 to 2019. In 2018, he was co-funder of the spin-off ZenaByte s.r.l. In 2019, he obtained the Italian National Scientific Qualification for the role of Full Professor in Computer Engineering. He is currently an Associate Professor in Computer Science at University of Pisa. His first main topic of research is the Statistical Learning Theory with particular focus on the theoretical aspects of the problems of (Semi) Supervised Model Selection and Error Estimation. His second main topic of research is Data Science with particular reference to the solution of real world problems by exploiting and improving the most recent Learning Algorithms and Theoretical Results in the fields of Machine Learning and Data Mining.

Chapter 1
Introduction



How can we select the best performing data-driven model and quantify its generalization error? This question has received a solid answer from the field of statistical inference since the last century and before [1, 2].

The now classic approach of parametric statistics [3–5] identifies a family of models (e.g. linear functions), a noise assumption (e.g. Gaussian) and, given some data, easily provides a criteria for choosing the best model, along with a quantification of the uncertainty or, in modern terms, an estimation of the generalization error in terms of a confidence interval. On the contrary, non-parametric statistics addresses the problem of deriving all the information directly from the data, without any assumption on the model family nor any other information that is external to the data set itself [6, 7].

With the advent of the digital information era, this approach has gained more and more popularity, up to the point of suggesting that effective data-driven models, with the desired accuracy, can be generated by simply collecting more and more data (see the work of Dhar [8] for some insights on this provocative and inexact but, unfortunately, widespread belief).

However, is it really possible to perform statistical inference for building predictive models without any assumption? Unfortunately, the series of no-free-lunch theorems provided a negative answer to this question [9]. They also showed that, in general, is not even possible to solve apparently simpler problems, like differentiating noise from data, no matter how large the data set is [10].

SLT addresses exactly this problem, by trying to find necessary and sufficient conditions for non-parametric inference to build data-driven models from data or, using the language of SLT, learn an optimal model from data [11]. The main SLT results have been obtained by deriving non-asymptotic bounds on the generalization error of a model or, to be more precise, upper and lower bounds on the excess risk between the optimal predictor and the learned model, as a function of the, possibly infinite and unknown, family of models and the number of available samples [11].

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L. Oneto, *Model Selection and Error Estimation in a Nutshell*,
Modeling and Optimization in Science and Technologies 15,
https://doi.org/10.1007/978-3-030-24359-3_1

An important byproduct of SLT has been the (theoretical) possibility of applying these bounds for solving the problems raised by our first question, about the quality and the performance of the learned model. However, for a long time, SLT has been considered only a theoretical, albeit very sound and deep, statistical framework, without any real applicability to practical problems [12]. Only in the last decade, with important advances in this field, it has been shown that SLT can provide practical answers [13–18].

We review here the main results of SLT for the purpose to select the best performing data-driven model and to quantify its generalization error. Note that we will not cover all the approaches available in the literature, since it would be almost impossible and also not very useful, but we will cover only the ground-breaking results in the field since all the other approaches can be seen as a small modification or combination of these ground-breaking achievements. Inspired by the idea of Shewchuk [19], our purpose is to provide an intelligible overview of the ideas, the hypotheses, the advantages and the disadvantages of the different approaches developed in the SLT framework. SLT is still an open field of research but can be the starting point for a better understanding of the methodologies able to rigorously assess the performance and reliability of data-driven models.

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Chapter 2
The "Five W" of MS and EE



Before starting with the technical parts we would like to answer five simple questions regarding the problem of MS and EE:

- **What** is MS and EE?
- **Why** should we care?
- **Who** should care?
- **Where** and **When** this problem has been addressed?

For what concerns the first "W", MS and EE can be defined respectively as the problem of selecting the data-driven model with the highest accuracy and the problem of estimating the error that the selected model will exhibit on previously unseen data by relying only on the available data. Example of MS problems are: choosing between different learning algorithms (e.g. Support Vector Machines [1], Random Forests [2], Neural Networks [3], Gaussian Processes [4], Nearest Neighbor [5], and Decision Trees [6]), setting the hyperparameters of a learning algorithm (e.g. the regularization in Support Vector Machines, the number of layers and neurons per layer in a Neural Network, the depth of a Decision Tree, and k in k-Nearest Neighbor), and choosing the structure of a learning algorithm (e.g. the type of regularizer in Regularize Least Squares [7, 8] and the families of kernels in Multiple Kernel Learning [9]). Once all the choices have been performed during the MS phase and the data-driven model has been built, the EE phase deals with the problem of estimating the error that this model will exibit on previously unseen data based on different tools such as: probability inequalities (e.g. Hoeffding Inequalities [10], Clopper-Pearson Inequalities [11]), concentration inequalities (e.g. Bounded Difference Function [12], Self Bounding Functions [13–16]), and moment inequalities [17]. Note that, in this work, we will deal only with methods which do not make any assumption on the noise that corrupts the data and that require only quantities that can be computed on the data themselves.

Regarding the second "W", MS and EE is a fundamental issue when it comes built data driven models. The first reason is that even advanced practitioners and researchers in the field often fail to perform MS and EE in the correct way. A quite

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L. Oneto, *Model Selection and Error Estimation in a Nutshell*,
Modeling and Optimization in Science and Technologies 15,
https://doi.org/10.1007/978-3-030-24359-3_2

representative evidence of this fact is a recent work published in one of the best journals in the field [18] where the performance of the state-of-the-art learning algorithms have been compared on a large series of benchmark datasets. In this work it is stated that Random Forests is the best performing algorithm but in a recent work [19] other researchers showed that the original work [18] contained a pretty serious flaws in how the MS and EE have been performed: in particular, an hold out dataset was missing leading to biased results. The second reason is rather simple: any data scientist wants to build the best data-driven model in terms of accuracy and have an idea of its actual performances when the model will be exploited in a real world environment. The third reason is that a series of no-free-lunch theorems [20, 21] ensure us that MS and EE will be always necessary since there will never exist a golden learning algorithm able to solve all the data related problems in the optimal way. The last, but not less important, reason is that in literature, due to the lack of knowledge of the problem of MS and EE, most of the research findings are false or report biased results [22, 23].

Regarding the "Who", the answer is a simple consequence of the "Why" since every data scientist, machine learner, data miner, and every practitioner which uses data should understand these concepts in order to provide reliable analyses, results, insights, and models. Probably the most technical issues of MS and EE are not fundamental for practitioners but having a general idea of the problems and the state-of-the-art tools and techniques able to address them is crucial for everyone in order to obtain rigorous and unbiased results.

Finally for the "Where" and "When" the answer, is simpler. The problem of MS and EE have been addressed in very theoretical Machine Learning and advanced Statistics papers mainly from the 1960 until today. Because of its intrinsic theoretical foundations it is prohibitive for a practitioner to read and comprehend the notation and the ideas proposed in these papers. This book is born from this observation and from a quite inspiring work on the gradient descend algorithm [24]. In fact, most of the time, the ideas behind the methods are quite simple and easy to apply but the technicalities and the knowledge taken for granted make most of the available literature hard to comprehend. In this book we will try to keep the presentation as simple as possible by presenting the problems, the hypotheses, and the ideas behind the methods without going in too many technical details but still presenting the state-of-the-art results in the field. In particular, we will start from the first works of 1960 about probability inequalities [10, 11, 25–40], and proceed with the asymptotic analysis [1, 41–43] of 1970, and concentration inequalities [13–17, 44–48] of 1980, then move to the finite sample analysis [49–103] of 1990, the milestone results of the 2000 about learnability [31, 104, 104–111], until the most recent results of 2010 on interactive data analysis [22, 112–126].